

Impacts of Structural Properties of Myosin II Filaments on Force Generation

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The authors present a **useful** agent-based model to study the tensile force generated by myosin mini-filaments in actin systems (bundles and networks); by numerically solving a mechanical model of myosin-II filaments, the authors provide insights into how the geometry of the molecular components and their elastic responses determine the force production. This work is of interest to biophysicists (in particular theoreticians) investigating force generation of motor molecules from a biomechanical engineering and physics perspective. The authors **convincingly** show that cooperative effects between multiple myosin filaments can enhance the total force generated, but not the efficiency of force generation (force per myosin) if passive cross-linkers are present. This work would benefit from a more extensive discussion of the physiological relevance of the results in view of the existing experimental literature, and how the principles that govern the behavior could be different for different motor proteins.

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Abstract

Cells need intracellular forces for their physiological functions, such as migration, cytokinesis, and morphogenesis. The actin cytoskeleton generates a large fraction of the forces via interactions between cytoskeletal components, such as actin filament (F-actin), myosin, and actin cross-linking proteins (ACPs). Myosin II plays the most important role in cellular force generation. Myosin II molecules self-assemble into filaments with different structures depending on myosin II isoforms and other conditions such as pH and ionic concentration. It has remained elusive how force generation in actomyosin structures is affected by the architecture of myosin II filaments. In this study, we employed an agent-based model to investigate the effects of the structural properties of myosin II filaments on force generation in disorganized actomyosin structures. We demonstrated that the magnitude of forces and the efficiency of force generation can vary over a wide range depending on the number and spatial distribution of myosin II filaments. Further, we showed that the number

of myosin heads and the length of a bare zone at the center of myosin II filaments without heads highly affect the force generation process in bundles and networks. Our study provides insights into understanding the roles of the structural properties of myosin II filaments in actomyosin contractility.

Introduction

Cells require forces for a wide variety of physiological functions, including cytokinesis, migration, and morphogenesis [1]. It is well-known that mechanical forces are produced mainly by molecular interactions between filamentous actin (F-actin) and myosin II motor proteins in the actin cytoskeleton [2]. Myosin II motors walk toward the barbed end of F-actin using chemical energy stored in adenosine triphosphate (ATP). Myosin II molecules consist of two heads with a long tail, and they self-assemble into filamentous structures [3]. There are three isoforms of myosin II: non-muscle, smooth muscle, and skeletal muscle myosins [4]. These myosin isoforms form different filamentous structures whose length ranges from $\sim 0.3 \mu\text{m}$ to $\sim 1.5 \mu\text{m}$ with ~ 56 to ~ 800 heads [5–12]. Unlike smooth muscle myosin forming a side-polar filament, non-muscle and skeletal muscle myosins form a bipolar filament with two sets of myosin heads located at both ends of the filament and a bare zone at its center whose length is $\sim 160 \text{ nm}$. Although many studies have reported that the center bare zone provides binding sites for multiple molecules, its contributions to the force generation remain elusive. The number of myosin II molecules in the myosin filament also varies depending on conditions, such as pH level and ionic concentration [13–16]. Individual myosin II heads have a relatively low duty ratio, meaning that they spend only a small fraction of their lifetime in the bound state unlike processive (i.e., high duty ratio) motors, such as myosin V or kinesin [17]. Thus, heads in a single myosin II molecule cannot walk along F-actin over a long distance by themselves. Myosin II circumvents this issue by forming the filamentous structure with multiple heads. If a myosin II filament has a sufficient number of heads, there are always a few myosin heads bound to F-actin, so the myosin II filament can remain proximal to the F-actin.

Heads with opposite polarities in the myosin II filament pull F-actins in opposite directions by walking toward the barbed ends of those F-actins, developing both tensile and compressive forces in disorganized actomyosin structures which are different from sarcomere found in muscle cells [18]. Theoretical and computational studies demonstrated that F-actin is easily buckled by compressive forces, leaving net tensile forces in disorganized bundles or networks that can mediate diverse contractile behaviors [19–21]. Various reconstituted actomyosin systems have been employed to illuminate how contraction and force generation emerge from interactions between F-actins, myosin II filaments, and actin cross-linking proteins (ACPs) [22–27]. Although they provided valuable insights, it has been understood poorly how various structural properties of myosin II filaments affect contraction or force generation in disorganized actomyosin structures.

Several computational models have been used to study the actomyosin contractility [28–34]. However, there were several limitations in the models. For example, some of the models treated myosin II filaments as either points [34, 35], rods only with two binding sites [31, 36, 37], or force dipoles [38]. Such drastically simplified structures significantly differ from the real structure of the myosin II filament. In this study, we used our well-established agent-based model with detailed descriptions of the structure of myosin bipolar filaments [20, 39–43], to illuminate how the number, length, bare zone size, and spatial distribution of myosin bipolar filaments influence force generation in disorganized actomyosin bundles and networks.

Materials and methods

Model overview

The detailed explanations about our model and all parameters used in the model are explained in Supplementary Text and Supplementary Tables. Our model consists of only three key cytoskeletal elements – F-actin, motor, and ACPs – among >100 proteins found in the actomyosin structures of cells [44]. They are simplified via cylindrical segments (**Fig. 1A**); F-actin is represented by serially connected segments with 140 nm in length. Each ACP comprises two arms with 23.5 nm in length connected to its center point. To mimic the structure of bipolar filaments, each motor has a backbone, consisting of serially linked segments, and two arms on each endpoint of the backbone segments that represent 8 myosin heads ($N_h = 8$). The reference length of backbone segments (L_{MB}) is 42 nm. The displacements of all the cylindrical segments at each time step are calculated by the Langevin equation and the forward Euler integration scheme. Deterministic forces in the Langevin equation include extensional and bending forces that maintain the equilibrium lengths of segments and equilibrium angles formed by segments, respectively, as well as a repulsive force exerted on overlapping pairs of actin segments for considering volume-exclusion effects.

Using the model, we simulate three types of systems: two filaments, bundles, and networks. In the two-filament simulations, a pair of anti-parallel F-actins with 19 μm in length are allocated in a rectangular computational domain ($5 \times 5 \times 20 \mu\text{m}$) with the periodic boundary condition (PBC) in x and y directions and the repulsive boundary condition in z direction. The barbed end of F-actins is clamped to two finite boundaries normal to the z direction. In the bundle simulations, we employ the same rectangular domain with the PBC in all directions (**Fig. 1B**). As in our previous study [42], we allocate F-actins in specific x and y coordinates. The number of possible positions for the allocation in x or y direction is defined by N_F , and spacing between adjacent coordinates in each direction is set to 27 nm. In each set of x and y positions, we position two F-actins with 9 μm in length in random z position with random polarity. Thus, the total number of F-actins in the bundle is $2N_F^2$. In addition, the network simulations are conducted using a thin rectangular computational domain ($20 \times 20 \times 0.1 \mu\text{m}$) with the PBC in x and y directions and the repulsive boundary condition in z direction (**Fig. 1C**). In the domain, F-actins with $\sim 10 \mu\text{m}$ in average length are allocated randomly in terms of positions and orientations as explained in Supplementary Text in detail. At the beginning of all simulations, while F-actins remain stationary, ACPs bind to F-actins to form permanent cross-linking points, and the arms of motors formed via the self-assembly of backbone segments bind to F-actins. After that, F-actins are allowed to move, and motor arms start walking and unbinding at force-dependent rates.

Variations in motor structures

In our study, we vary three structural properties of motors: the number of arms per motor (N_a), the length of the bare zone (L_{bz}), and spacing between motor arms (L_{sp}). We change the size of motors in three different ways. In the first way, N_a is increased by connecting more segments with arms for a backbone, whereas L_{bz} and L_{sp} are equal to L_{MB} (**Fig. 1E, i**). In the second way, L_{bz} is increased by adding more segments without arms to the bare zone, whereas N_a is unchanged, and L_{sp} is equal to L_{MB} (**Fig. 1E, ii**). In the third way, L_{sp} and L_{bz} are increased by making L_{MB} higher, whereas N_a is unchanged (**Fig. 1E, iii**).

Measurement of contractile forces

In the two-filament simulations and the bundle simulations, we measure tensile forces generated by the systems as follows. First, all segments crossing the cross-sections of the domain located every 200 nm in the z direction, which can be for either of F-actin, motor arm, motor backbone, or ACP arm, are identified (**Fig. S1**). Then, the z component of spring forces acting on those

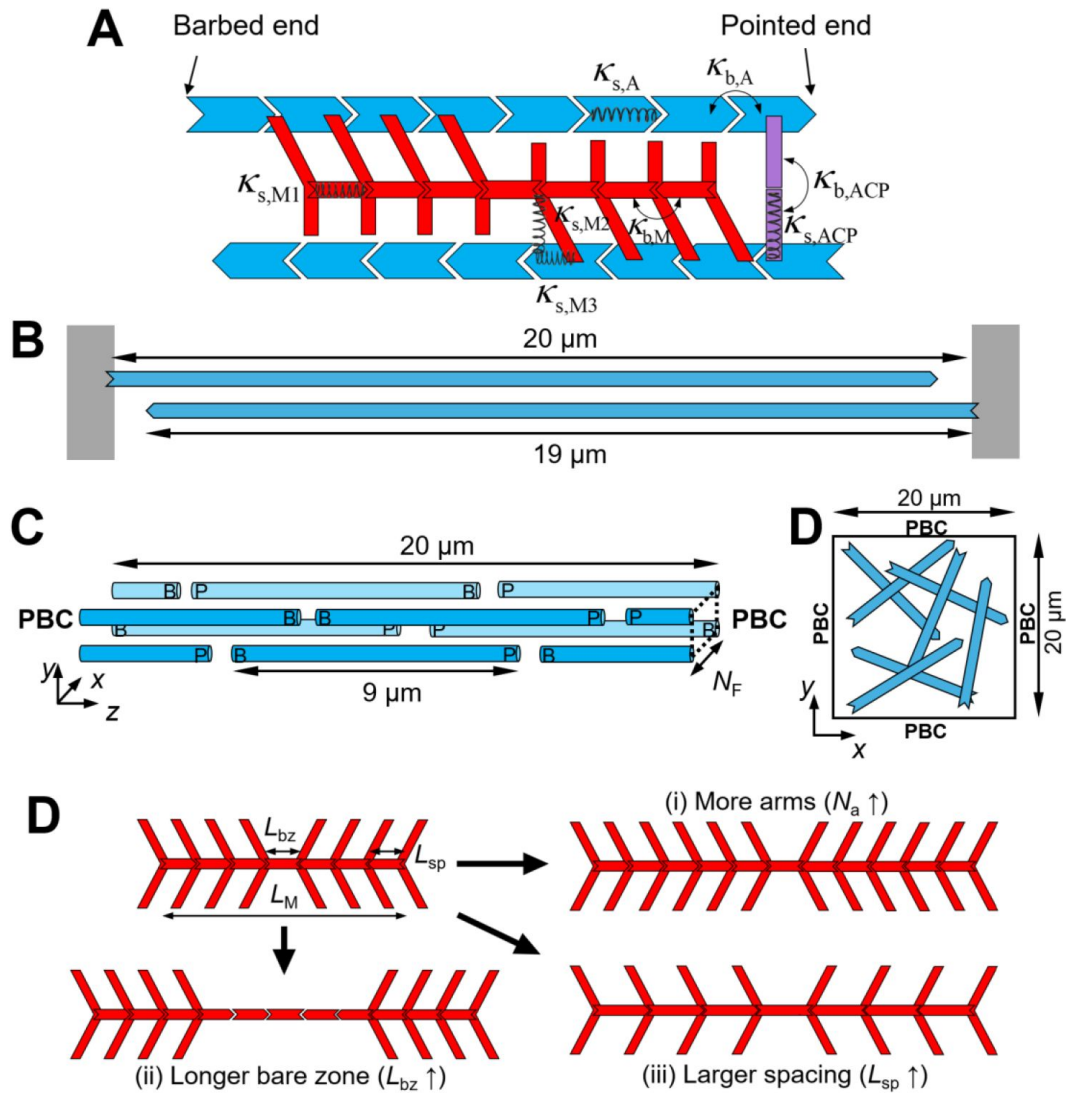


Fig. 1.

Modeling setups.

(A) In the model, F-actin (blue), actin cross-linking protein (ACP, purple), and motor (red) are simplified by cylindrical segments. F-actin has polarity defined by barbed and pointed ends. Motors consist of a backbone with motor arms that can bind to and walk along F-actin. ACPs comprise two segments connected at the center point. κ_s and κ_b represent extensional and bending stiffnesses, respectively. (B) Two-filament system consisting of two F-actins whose barbed ends are clamped to rigid boundaries (gray). (C) The disorganized bundle system with $2N_F^2$ F-actins randomly located and oriented in the presence of the periodic boundary condition (PBC) in the z direction, where N_F is a parameter defining bundle thickness. (D) The two-dimensional network system consisting of F-actins with random positions and orientations with the PBC in x and y directions. (E) A variation in the motor structure in three different ways: (i) increasing the number of motor arms ($N_a \uparrow$), (ii) increasing the bare zone length ($L_{bz} \uparrow$), and (iii) increasing a spacing between motor arms ($L_{sp} \uparrow$).

segments is summed. Using these sums, four curves can be drawn as a function of z to show which segment supports more or less tensile loads in each z position. Note that at equilibrium, the sum of four forces is very similar regardless of z position, meaning that a large fraction of forces developed by motors act as spring forces rather than bending forces. The average of the sums calculated on all cross-sections at a steady state is considered the total force acting on the system, F_{tot} . In the network simulations, tensile forces are measured in a similar manner, using 20 cross-sections of the domain located evenly in the x and y directions. x or y component of tensile forces acting on all segments crossing each cross-section is summed at a steady state. The average of sums calculated on 40 cross-sections is considered F_{tot} .

Results

The distribution of motors and ACPs plays a key role in force generation

A previous in vitro study reported that tension developed in a thin bundle was almost directly proportional to the number of myosin heads in each motor, not to the number of motors [4]. Although their experiments did not include any ACP, their theoretical explanation was based on an assumption that the bundle consists of serially connected contractile units like sarcomeres in muscle cells [4]. They admitted that these contractile units do not have structural analogs in their disordered actomyosin bundles.

To verify their hypothesis, we first used a simple minimal model composed of two anti-parallel F-actins, two motors, and 16 ACPs (Figs. 1A and 2). The number of arms per motor was set to $N_a = 24$. We ran 20 simulations with random z positions of motors and ACPs and found that the total force generated by the system, F_{tot} , fell into the following range (Fig. 2A):

$$0.5F_M^{\max} \leq F_{\text{tot}} \leq F_M^{\max}. \quad (1)$$

Where $F_M^{\max}$ is the maximal force that all motors can generate in this system:

$$F_M^{\max} = \frac{1}{2} \sum_i^{N_M} F_{M,z}^i \quad (2)$$

Where $F_{M,z}^i$ is the z component of spring forces exerted by i th motor at a steady state, and N_M is the number of motors. In this example, N_M is 2. Note that only a quarter of motor arms (i.e., $N_a / 4$) can bind to one F-actin due to two constraints assumed for the binding of the motor arms: motor arms can bind to F-actin when the arms are properly aligned with the polarity of F-actin, and two arms connected to the same point on a backbone cannot bind to the same F-actin. With two antiparallel F-actins, up to half of the motor arms can stay in the bound state. Thus, $F_{M,z}^i$ of a motor bound to two anti-parallel F-actins at the steady state is close to $F_{\text{st}}N_hN_a/2$, where F_{st} is the stall force of one myosin head, and N_h indicates the number of myosin heads represented by each motor arm. We found that a difference in F_{tot} originated from the relative positions of motors and ACPs. When two motors were located in a very similar position with an almost full overlap by chance, F_{tot} was close to $F_M^{\max}$. When the two motors stayed apart without any overlap, F_{tot} was close to either $0.5F_M^{\max}$ or $F_M^{\max}$ depending on whether or not there were ACPs between them. In the absence of ACPs between two motors, forces generated by the arms of two motors could add up to develop larger tensile forces on F-actins (Figs. 2B, E). However, when ACPs existed between the two motors, it resulted in the buckling of F-actin between the ACPs and motor arms on one side, and counterbalanced the tension generated by motor arms on the other side (Figs. 2C, F). As a result, F-actins ended up with feeling tension corresponding to a force generated by one motor. These ACPs between two motors divide the bundle into serially connected contractile units

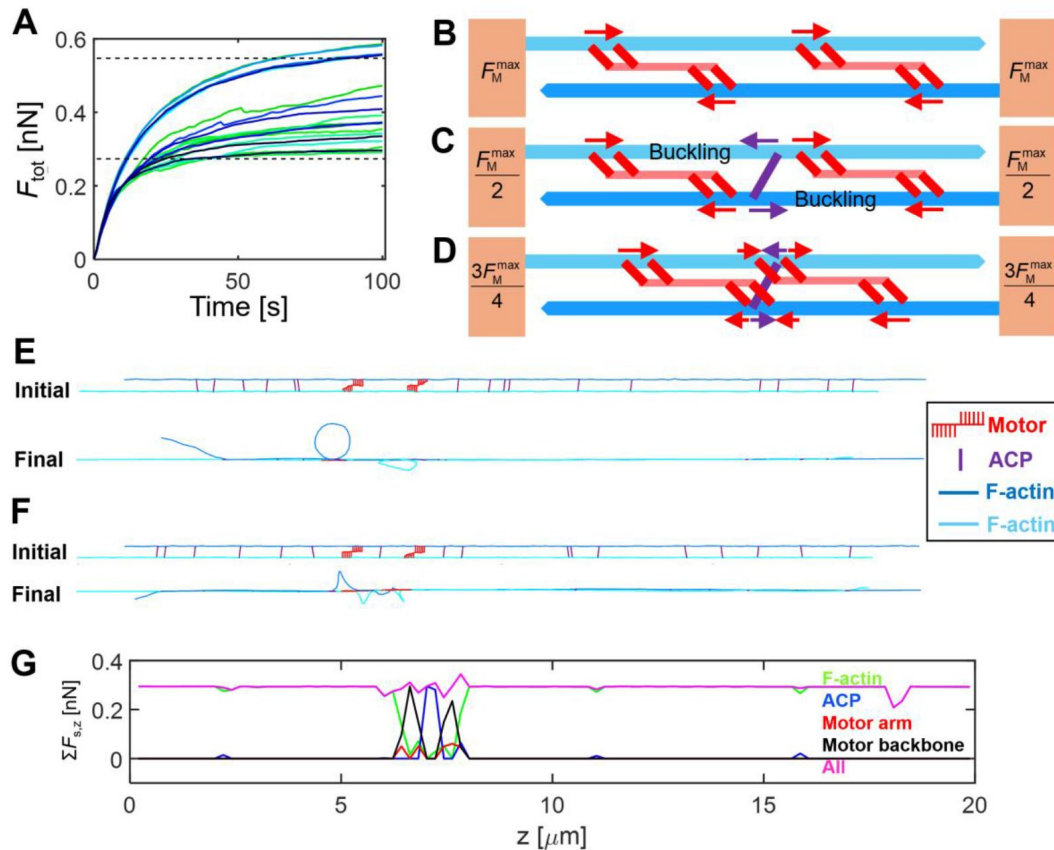


Fig. 2.

Interactions between motors and ACPs regulate force generation.

(A) Time evolution of the force generated by two motors in the two-filament system. The upper and lower dashed lines indicate ideal upper and lower limits of a force that two motors can generate, respectively. Different colors represent distinct cases repeated 20 times. (B) Without any ACP between two motors, they can generate a force close to the upper limit which is two-fold larger than a force that one motor can generate ($= F_M^{\max}$). (C) With ACP(s) between two motors, ACPs counterbalance a force generated by one of the motors. Thus, they can generate a force close to the lower limit which is equal to the force that one motor can generate ($= F_M^{\max}/2$). Part of F-actins is buckled due to two forces with opposite directions. (D) If two motors are close to each other, ACP can counterbalance a fraction of the force generated by one motor. Then, two motors can generate a force between the upper and lower limits. (E, F) Initial and final configurations (E) without or (F) with ACPs between two motors. The vertical dimension is increased 10 times to show the configurations clearly. (G) Measurement of tensile forces acting on F-actins (green), ACPs (blue), motor arms (red), motor backbones (black), or all (magenta) in z direction.

as the assumption of the theoretical explanation in the other study mentioned earlier [44, 45]. These ACPs were observed to experience larger tension than the rest of the ACPs since they directly counterbalanced large tension generated by motors (Fig. 2G). There were quite a few cases showing intermediate level of bundle tension between $0.5F_M^{\max}$ and $F_M^{\max}$ (Fig. 2A). These cases had motors with a partial overlap and ACPs located within the z range spanned by one of these motors. Due to the partial overlap with ACPs, forces generated by two motors partially added up, resulting in the intermediate tension (Fig. 2D).

Force generation in disorganized bundles is also regulated by the same mechanism

To verify the importance of the spatial distributions of motors and ACPs for force generation in more physiologically relevant structures, we repeated simulations using disorganized bundles with various sizes between $N_F = 2$ (8 filaments) and $N_F = 7$ (98 filaments) (Figs. 1B and 3A). The densities of motors and ACPs were fixed at $R_{ACP} = 0.04$ and $R_M = 0.005$, resulting in N_M ranging between 4 and 52. In all cases, each motor still had 24 arms ($N_a = 24$). When the bundle became thicker by increasing N_F , a tensile force acting on the bundle (F_{tot}) also increased (Fig. 3B). Because the motor density was fixed, more motors were present in thicker bundles, generating larger F_{tot} . In case of the thickest bundle ($N_F = 7$), actin concentration was 12.25-fold higher ($= 98 / 8$) compared to the thinnest bundle ($N_F = 2$), so there were 13-fold more motors.

We defined the efficiency of force generation:

$$\eta = \frac{F_{tot}}{F_M^{\max}} \quad (3)$$

In these bundles, $F_M^{\max}$ is still defined by Eq. 2, but $F_{M,z}^i$ was close to $F_{st}N_hN_a$ because there were more than one F-actin to bind for each polarity. Interestingly, η was quite similar in cases with $N_F = 4, 6$, and 7 (Fig. 3C). By contrast, cases with $N_F = 2$ exhibited much higher η than the other cases. We probed why motors could generate forces in the thinnest bundle in a more efficient manner. With $N_F = 2$, there were only 4 motors. The length of each motor with 24 arms is 462 nm ($= 42 \text{ nm} \times 11$). The sum of the length of all 4 motors is only $\sim 9\%$ of bundle length, 20 μm , so they were less likely to significantly overlap with each other. Instead, most of them stayed apart with various distances. Given high ACP density, there were more than one ACP between adjacent motors. Thus, these motors formed separate contractile units, and their forces could not add up. Thus, η for the thinnest bundle was roughly $1/N_M$ because F_{tot} was close to the force generated by a single motor. Indeed, η for the thinnest bundle was close to 0.25, confirming our rationale. If motors behave in the same manner in thicker bundles ($N_F > 2$), η will be smaller than 0.25 because N_M is proportional to N_F^2 due to fixed motor density. Although η was actually smaller than 0.25 in thicker bundles, it was ~ 0.12 in all cases with $N_F > 2$, which was larger than the prediction, $1/N_M$. As the bundle was thicker, there was a higher chance for motors to overlap with each other or stay closely due to higher N_M . Then, some of these proximal motors could add up their forces if there is no ACP between them as discussed earlier. If the bundle has any set of such “cooperative” motors, F_{tot} can become larger than a maximal force generated by a single motor since F_{tot} is determined by the largest force generated from one of the contractile units. As N_M increases due to high N_F , the number of the cooperative motors tends to increase, resulting in larger F_{tot} . Thus, as N_F increases, both F_{tot} and $F_M^{\max}$ increase, so η can be higher than $1/N_M$ and rather insensitive to a change in N_F when N_F is not small.

Bundles with more, localized motors generate larger forces

Based on prior observations, overlaps between motors highly affect the force generation process. The extent of overlaps is determined by N_M if the bundle length is fixed. To understand how force generation in the bundles depends on N_M more systematically, we ran simulations with a wide

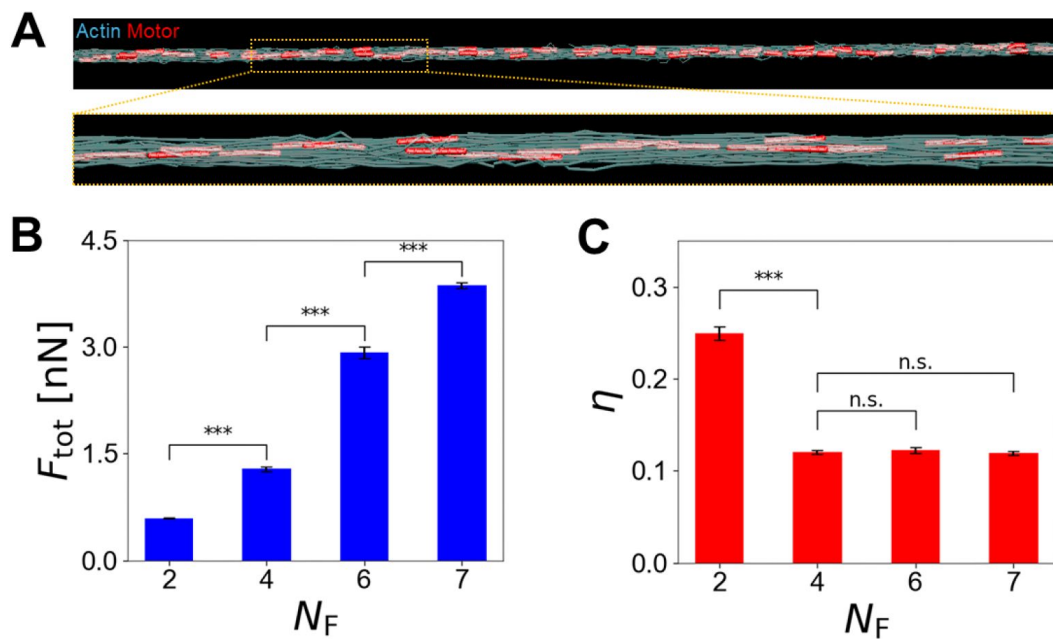


Fig. 3.

With fixed motor density, thicker bundles generate larger force in a less efficient manner.

(A) An example of disorganized bundles with $N_F = 7$ visualized at the beginning of the simulation, where N_F is a parameter defining bundle thickness. F-actins are visualized as transparent elements to show the positions of motors. (B) Bundle-level force and (C) the efficiency of force generation measured at a steady state with different N_F . In thicker bundles ($N_F > 2$), larger forces were generated, but the efficiency was lower than that in the thinnest bundle ($N_F = 2$). *** represents $p \leq 0.001$, and n.s. represents $p > 0.05$.

range of N_M , using the thickest bundle ($N_F = 7$) and the same $R_{ACP} = 0.04$ and $N_a = 24$. The bundle tension, F_{tot} , showed a tendency to increase with higher N_M , but η was inversely proportional to N_M (Fig. 4A). With low N_M , F_{tot} was less sensitive to an increase in N_M , whereas η was very sensitive to the increase in N_M . If there are a small number of motors in the bundle, adding a few more motors would not lead to a significant increase in F_{tot} because the new motors are not likely to be located in positions where they can add up forces with other motors (Fig. 4B). Instead, they will merely increase the number of contractile units, most of which have only one motor. Thus, as explained in Eq. 3, η was close to $1/N_M$. With higher N_M , F_{tot} was sensitive to a change in N_M because adding more motors to the bundle contributes to an increase in the number of motors in each contractile unit (Fig. 4B). When N_M was very high, F_{tot} was almost directly proportional to N_M with slope ~ 1 since all parts of the bundle were already occupied by motors, so adding more motors caused a direct impact on the magnitude of F_{tot} . In the range of high N_M , η was much less sensitive to a change in N_M . For example, when N_M was varied from 10 to 1,000, η was reduced to approximately half. The insensitivity of η is attributed to an increase in both F_{tot} and F_M^{max} with higher N_M as explained earlier. The plateau level of η at ~ 0.08 is related to the minimum number of motors required for saturating an entire bundle, implying that the plateau level would be higher if each motor is longer.

To verify this rationale, we developed a way to estimate the maximum number of overlapping motors, $\mathcal{E}$, using our simulation data:

$$\mathcal{E} = \max_{\substack{i=1 \dots N_M \\ j=L \text{ or } R}} \left(1 + \sum_{k \neq i}^{N_M} \xi_{ik}^j \right) \quad (4)$$

where ξ_{ik}^j is:

$$\xi_{ik}^j = \begin{cases} 1 & \text{if } L_c \leq L_{ov} \leq L_M \\ L_{ov} / L_c & \text{if } 0 < L_{ov} < L_c \\ 0 & \text{if } L_{ov} = 0 \end{cases} \quad (5)$$

i and k are the indices of motors, j denotes the left or right side of a motor, and L_{ov} is an overlap distance between two motors. In addition, L_c is the critical length required for the cooperative overlap:

$$L_c = 2L_{sp} \left(\frac{N_a}{4} - 1 \right) \quad (6)$$

L_c is two-fold greater than the length occupied by motor arms on one side of a motor backbone. Two motors are considered a cooperative motor pair if L_{ov} is equal to or greater than L_c (Fig. S2). With this cooperative overlap, ACPs cannot counterbalance forces exerted by motor arms. If L_{ov} is shorter than L_c and but greater than zero, two motors are considered as a partially cooperative motor pair. By comparing all the motor combinations, $\mathcal{E}$ can be obtained. $\mathcal{E}$ can range between 1 (no overlap between motors) and N_M (cooperative overlap between all motors). Using this $\mathcal{E}$, a force acting on the bundle can be estimated as F_{est} :

$$F_{est} = \frac{F_{st} N_h N_a \mathcal{E}}{2} \quad (7)$$

The sensitivity of F_{est} to an increase of N_M is high at large N_M with a slope close to 1 (Fig. 4C), which is consistent with our rationale described earlier. F_{est} was generally smaller than F_{tot} because this analysis does not account for actual bundle geometry consisting of multiple F-actins;

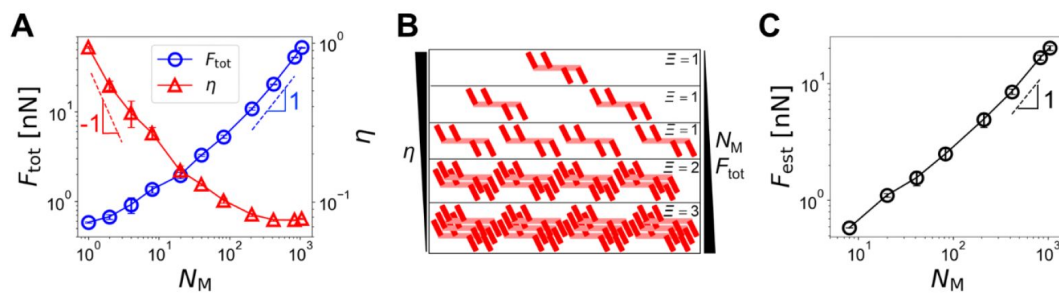


Fig. 4.

An increase in the number of motors (N_M) in disorganized bundles results in larger forces but smaller force generation efficiency.

The thickest bundle ($N_F = 7$) was used for all cases. (A) Bundle-level force (blue circles) and the efficiency of force generation (red triangles) with a wide range of N_M between 1 and 1045. With more motors, a larger force (F_{tot}) was generated, but the efficiency (η) was lower. (B) Configuration of motors with different N_M . Ξ represents the estimated number of motors in the strongest contractile unit. With small N_M , F_{tot} and Ξ are unlikely to increase significantly until an entire bundle is occupied by motors, so η is roughly $1/N_M$. By contrast, with high N_M , an increase N_M directly enhances F_{tot} and Ξ , and η almost remains constant. (C) Prediction of a force using the positions of motors. To find the estimated force (F_{est}), Ξ is calculated first, and Eq. 7 is used.

if two motors are located far from each other in x or y direction, they may not counterbalance or add up forces. Nevertheless, we found that F_{est} captures the overall dependence of F_{tot} on parameters well.

So far, we have assumed that motors could be randomly located at any part of the bundle. If motors are confined within a smaller region, the extent of overlaps between motors (i.e., $\mathcal{E}$) could be increased with the same N_M , enhancing force generation. We tested the effects of motor distribution with $N_F = 7$, $R_{\text{ACP}} = 0.04$, $N_M = 52$, and $N_a = 24$. We allowed motors to be located only within a portion of the bundle defined by f between 0 and 1. $f = 1$ means that motors can be located anywhere. The center of the region for motor allocation is equal to the center of the bundle at $z = 10 \mu\text{m}$. It was observed that a bundle generated the largest F_{tot} , and η was the highest when motors were located near the center ($f = 0.06$) (Fig. 5A). Note that F_M^{max} in the denominator of Eq. 3 is identical in these cases, meaning that F_{tot} is directly proportional to η . Motors localized in relatively the same position are more likely to overlap in the cooperative manner (i.e., $L_{\text{ov}} \geq L_c$). Therefore, a large fraction of the motors were involved with the formation of a strong contractile unit (Fig. 5B). As f increased, motors were distributed on the bundle more sparsely. Then, many motors were partially overlapped or separated from each other. As a result, F_{tot} and η decreased (Fig. 5A). The effect of a variation in f on force generation was also reproduced well by the theoretical model explained earlier (Fig. 5C).

The structure of motors influences force generation in bundles

We have employed motors with 24 arms whose length is $L_M = 462 \text{ nm}$ with $L_{\text{sp}} = 42 \text{ nm}$ and $L_{\text{bz}} = 42 \text{ nm}$. As mentioned earlier, the structure of myosin thick filaments can vary significantly, depending on myosin isoform and conditions. If motors are longer and have more arms, they may generate higher bundle forces. To verify this hypothesis, we tested cases with different N_a between 4 and 48 and the thickest bundle ($N_F = 7$), with the total number of arms in the system, $N_a N_M$, fixed. With increasing N_a , L_M was increased from 42 nm to 966 nm, but N_M was decreased from 313 to 26 (Fig. 1E, i). When L_M was increased by higher N_a , F_{tot} increased (Fig. 6A, red). Because these cases had the same total number of motor arms, F_M^{max} in Eq. 3 was identical in all the cases. Thus, η showed the same tendency as F_{tot} (Fig. S3A); with longer motors, η was higher. The same tendency was observed in F_{est} (Fig. 6B, red). As motors have motor arms, individual contractile units become stronger, so F_{tot} and η are directly proportional to L_M if there is no overlap between motors (Fig. 6C). However, these motors are hard to overlap in a fully cooperative manner because L_c is large (Fig. 6D, top and Eqs. 5 and 6). Thus, $\mathcal{E}$ was actually smaller with higher L_M (Fig. S3B, red). This explains why the dependence of F_{tot} on L_M was weaker at high L_M than that at low L_M (Fig. 6A).

There are two additional ways to increase L_M without a change in N_a and N_M : increasing L_{bz} at the center of motors or increasing L_{sp} between motor arms uniformly (Fig. 1E, ii and iii). We ran simulations with a variation in either L_{sp} (between 42 nm and 138 nm) or L_{bz} (between 42 nm and 714 nm), using the thickest bundle ($N_F = 7$) with $N_a = 16$. We compared results from these new simulations with those obtained with a variation in N_a shown earlier. Interestingly, when L_M was similar, F_{tot} acquired with a change in L_{bz} was similar to that with a variation in N_a despite different N_M (Fig. 6A, blue), whereas F_{tot} in cases with a change in L_{sp} was noticeably lower (Fig. 6A, green). η showed identical tendency as F_{tot} because F_M^{max} was the same in all the cases (Fig. S3A). F_{est} also reproduced a similar tendency (Fig. 6B). If L_M is fixed, a longer bare zone makes motors have their arms near the two ends of their backbone. Then, motors can have a higher possibility to add up their forces because L_c is smaller (Fig. 6D, bottom and Eqs. 5 and 6). Thus, $\mathcal{E}$ was higher (Fig. S3B, blue), resulting in higher F_{tot} and η despite smaller N_a . By contrast, motors with uniform large spacing between arms need to have a more overlap to add up their forces (Fig. 6D, middle). Smaller $\mathcal{E}$ and N_a led to lower F_{tot} and η (Fig. S3B, green).

Fig. 5.

Motor distribution affects the force generation in disorganized bundles.

We varied the relative size of a region where motors were initially located, f . $f = 1$ means that motors can be located at any part of the bundle. (A) Bundle-level force (F_{tot}) and efficiency (η) depending on f . As motors were localized more closely to the center (i.e., smaller f), the force and the efficiency were higher. (B) Configuration of motors with different f . Ξ indicates the estimated number of motors in the strongest contractile unit. Given the number of motors, smaller f results in more cooperative overlaps (i.e., higher Ξ) and thus leads to higher F_{tot} and η . (C) Prediction of the bundle-level force with different f .

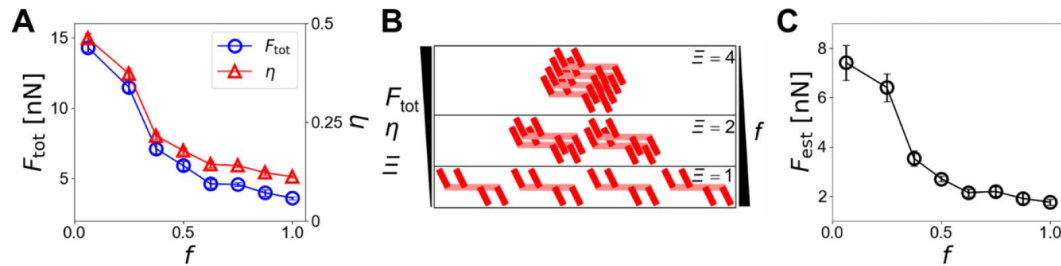
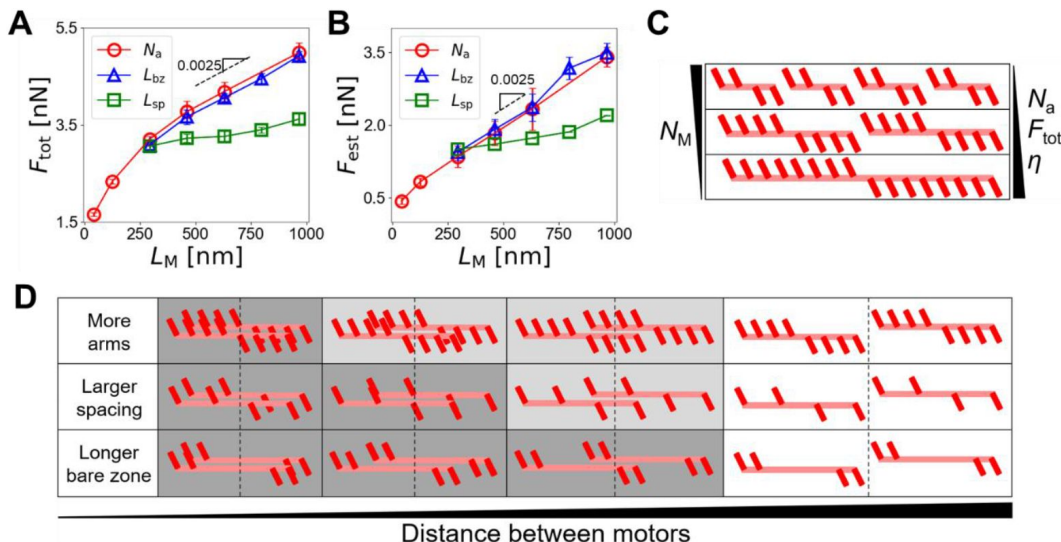


Fig. 6.

The architecture of motors impacts the force generation process in disorganized bundles.

(A) Bundle-level force (F_{tot}) depending on L_M . The motor length (L_M) is varied between 42 nm and 966 nm by changing either of the number of motor arms (N_a , red circles) between 4 and 48, the bare zone length (L_{bz} , blue triangles) between 42 nm and 714 nm, or the spacing of motor arms (L_{sp} , green squares) between 42 nm and 138 nm. (B) Prediction of the bundle-level force using the positions of motors. (C) As each motor has more arms ($N_a \uparrow$), F_{tot} and the efficiency of force generation (η) become higher because forces generated by motor arms are counterbalanced to a lesser extent. (D) Possible overlaps between two motors with different structures. A dark gray color indicates a fully cooperative overlap, and light gray indicates a partially cooperative overlap. To have the fully cooperative overlap, motors with many arms need to be located very closely, whereas motors with the long bare zone can overlap in the fully cooperative manner with a relatively long distance between them.



Force generation in actin networks is regulated by the same mechanism

To check the generality of our findings, we performed simulations using a two-dimensional (2D) network where F-actins and motors are randomly oriented without any bias, which is different from those in bundles (**Fig. 1C**). We varied either of N_M , N_a , L_{bz} , or L_{sp} as done for bundles (**Fig. 7A**). F_{tot} was smaller than the values measured in bundles under the same conditions due to the random orientations of motors (**Figs. 7B, C**).

Interestingly, we found that F_{tot} is proportional to $N_M^{0.65}$, which is close to $\sqrt{N_M}$. Note that F_{tot} was directly proportional to N_M at large N_M in case of the bundles (**Fig. 3A**). Considering that motors are uniformly distributed on a 2D network, this weaker dependence on N_M is expected; the average number of motors in each direction which can experience the cooperative overlap would be $\sim \sqrt{N_M}$. Maximal N_M tested with the network was $\sim 2,500$, so the dependence of F_{tot} on N_M with the network is similar to that with $N_M < \sim 50$ with the bundle (**Fig. 4A**). η showed similar magnitudes to those measured in the bundles (**Figs. 6B**, **S3**, and **S4**). Note that η was still calculated using **Eq. 3**, but F_M^{max} was obtained by summing the x or y component of forces exerted by all motors network and then averaging them:

$$F_M^{max} = \frac{1}{4} \left(\sum_i^{N_M} F_{M,x}^i + \sum_i^{N_M} F_{M,y}^i \right) \quad (8)$$

We found that the dependences of F_{tot} and η on these parameters are similar to those observed with bundles, meaning that force generation regulated by cooperative overlaps between motors takes place even in networks. Although forces exerted by motors are not oriented in the same direction in networks, these forces can be counterbalanced or add up depending on the relative positions of motors.

Discussion

Actomyosin contractility is well-conserved machinery for generating mechanical forces in animal cells, facilitating cytokinesis, cell migration, and tissue morphogenesis [18]. Myosin II, which is the most important molecular motor for cellular force generation, exists as a highly organized form called thick filaments. Myosin thick filaments have been studied extensively during recent decades. However, despite various forms of thick filaments found in different types of cells, the effects of their structural properties on force generation have not been understood well.

There was an in vitro study that employed thin disorganized actomyosin bundles consisting of a few F-actins with three myosin isoforms (skeletal muscle, smooth muscle, and non-muscle myosins) in order to investigate the effects of the size of myosin II filaments [4]. The study showed that a tensile force developed in bundles is directly proportional to the number of myosin heads in each myosin II filament if the total number of myosin heads is identical. They explained this direct proportionality via a theoretical bundle model consisting of serially-connected contractile units in which a single myosin II filament generates a tensile force. This is partially consistent with observations in our study. However, such explanation can be applied only to a bundle without any overlap between thick filaments, meaning that there are only a few thick filaments.

Although force generation by actomyosin contractility has been investigated in several theoretical and computational studies, most of the previous models used drastically simplified motors without consideration of thick filament structures [36, 37, 46]. Therefore, it was not feasible to

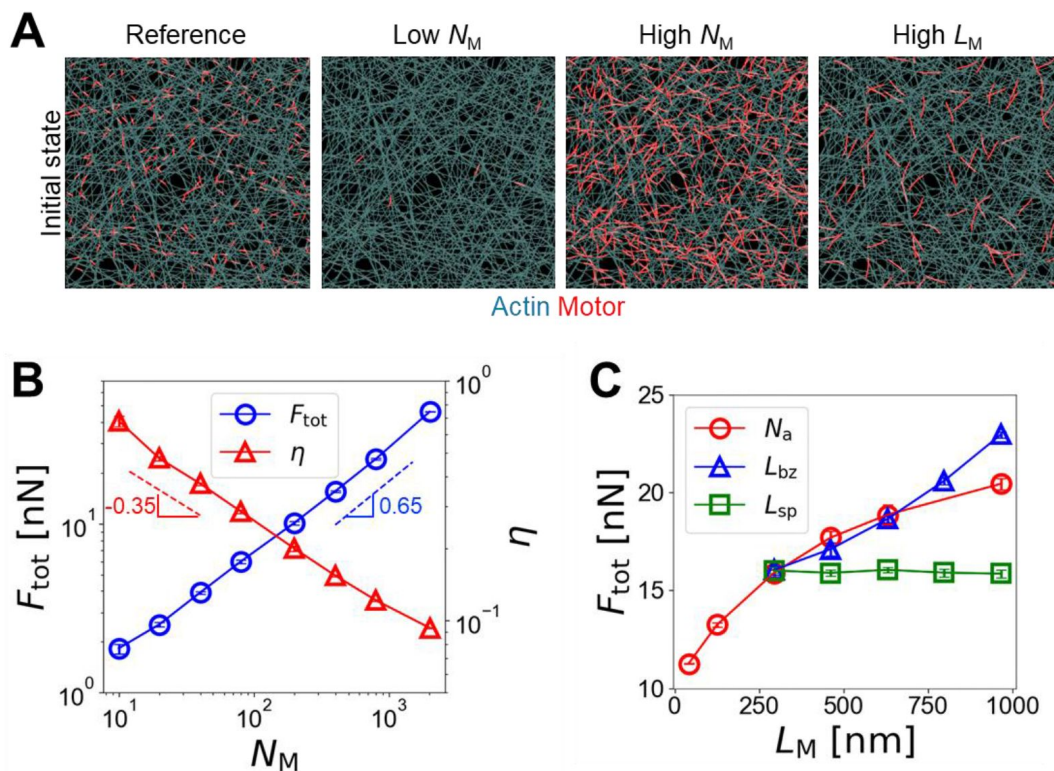


Fig. 7.

Force generation in two-dimensional actomyosin networks is governed by similar mechanisms.

(A) Examples of networks at the initial state under the reference condition, with a smaller or larger number of motors, and with longer motors. These snapshots show only a quarter of networks to better visualize individual motors. (B) Network-level tension (F_{tot}) and the efficiency of force generation (η) with a different number of motors between 10 and 2006. (C) F_{tot} depending on motor length varied by changing either of the number of motor arms (N_a , red circles) between 4 and 48, the bare zone length (L_{bz} , blue triangles) between 42 nm and 714 nm, or the spacing of motor arms (L_{sp} , green squares) between 42 nm and 138 nm.

investigate the importance of the structural properties of thick filaments in force generation in those studies. There were a few models accounting for thick filament structures. However, only one thick filament was simulated [32, 33] due to high computational cost, or force generation process was not probed [31]. In our study, using motors with the geometry of bipolar thick filaments, we systematically showed how the force generation process in disorganized actomyosin structures, including bundles and networks, is regulated by the number, spatial distribution, and structural properties of motors. First, using the simplest system with only two F-actins, we showed that motors with ACPs between them cannot add up their forces to generate a larger force (Fig. 2). Then, we probed the force generation process in disorganized bundles with different thickness and found that the efficiency of force generation was lower in thicker bundles although the bundle-level force was higher (Fig. 3). This was attributed to an increase in the number of motors in thicker bundles. We then found that a larger force was developed in the same bundle in a less efficient way when there were a larger number of motors (Fig. 4). When motors were sparsely distributed without an overlap between them, the efficiency of force generation was inversely proportional to the number of motors (i.e., $\eta \propto 1/N_M$). As the number of motors further increased, motors started cooperatively overlapping with each other, forming stronger contractile units. Thus, the efficiency became greater than $1/N_M$. We also tested the effects of motor distribution on force generation to verify the importance of cooperative overlaps (Fig. 5). As motors were distributed in a more confined region, bundles generated larger forces because there could be more cooperative overlaps between more densely distributed motors. We also found that force generation was enhanced and more efficient when motors were longer with more arms because forces generated by the arms of one motor can simply add up (Fig. 6). In addition, longer motors with a long bare zone and a few arms could generate large forces in bundles than longer motors with a small bare zone and large spacing between arms since the former can achieve the cooperative overlap by a much smaller overlap (Fig. 6). Our results imply that the consideration of F-actin connectivity may not be enough to accurately predict force generation unlike predictions from a recent study [46].

Although this study focused mainly on parameters related to motor structures, we expect that other parameters would affect the force generation process. For example, as we showed before [41, 42], a decrease in ACP density would reduce forces by deteriorating connectivity between filaments. With very low ACP density, some of neighboring motors may not have ACPs between them, thus adding up their forces as shown in Fig. 2. However, such low ACP density may not maintain the structure of bundles or cross-linked networks well. In addition, the force-dependent unbinding of ACPs could change the spatial distribution of ACPs during force generation. If they behave as a slip bond which unbinds more frequently with higher forces, ACPs may not stay between two motors for long time due to high tension [47]. Then, forces generated by two motors may have a higher chance to add up. By contrast, if they behave as a catch bond which unbinds less frequently with larger forces, more ACPs will be recruited between two motors, reducing a chance to add up forces. The length of actin filaments is unlikely to affect the force generation process significantly unless filaments are very short. Additionally, as we showed before [43, 48], actin turnover would reduce forces by competing with motor activities, change connectivity between filaments over time, and prevent motors from being stalled for long time, all of which could affect force generation.

Our findings can be applied to understand the structures of stress fibers that are known to mediate physiological processes, such as cell protrusion, cytokinesis, and cell shape maintenance [49, 50]. In stress fibers, non-muscle myosin II plays a main role in generating contractile forces. There are two types of contractile stress fibers: transverse arcs and ventral stress fibers [51]. Ventral stress fibers, which are assembled from pre-existing stress fiber precursors, have a sarcomere-like structure with cascaded (i.e., serially connected) contractile units divided by α -actinin which is one type of ACPs. By contrast, transverse arcs are smaller without distinct repeated structures with much fewer ACPs [51]. ACPs in ventral stress fibers can counterbalance forces generated by motors as shown in Fig. 2. The structure with cascaded

units and many ACPs prevents the bundle-generated force from increasing beyond a force generated by a single contractile unit. Therefore, it is expected that ventral stress fibers would generate smaller forces than transverse arcs if their thickness is similar. However, the bundle-generated force would be proportional to the thickness of bundles as shown in **Fig. 3**. Thus, ventral stress fibers, which are typically thicker, are known to generate larger tension than transverse arcs [52]. Although we focused on force generation, the contractile behaviors of actomyosin structures (i.e., a decrease in length) have also been of great interest. Our model can be used to study such contractile behaviors by deactivating the periodic boundary condition and removing connection between one end of bundle/network and a domain boundary as done previously [20]. To achieve higher contractile speed with the same total number of myosin heads, the existence of multiple contractile units would be better as suggested in a previous work [4]. This means that there is a trade-off between force generation and contractile speed. Previous studies also showed that the contractile speed of networks is proportional to motor density [18, 43, 53]. We may be able to use our model to systematically investigate how the contractile speed is regulated by parameters that we tested in this study, including the number, distribution, length, and structure of motors.

In conclusion, we probed how the force generation process in disorganized bundles and networks are regulated by the properties of myosin thick filaments. We found that more motors enable bundles and networks to generate larger tensile forces, but the efficiency of force generation was lower. With the same number of motors, forces generated by bundles and networks can vary to a large extent, depending on i) whether they are sparsely or densely distributed, ii) how many myosin heads each thick filament has, and iii) how long the bare zone at the center is. Our results can be partly verified using different myosin isoforms under different conditions as done in the previous study [4]. In addition, synthetic myosin thick filaments (i.e., one made with DNA origami [54]) whose structure can be designed artificially could be used to verify our findings.

Data availability

The source code of our model is available on GitHub: <https://github.com/ktyman2/ThickFilament>

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Supplementary materials

Brownian dynamics via the Langevin equation

In our model, F-actin consists of serially-connected cylindrical segments with barbed and pointed ends. Actin cross-linking proteins (ACPs) comprise two segments connected by an elastic hinge. Each motor mimics the structure of myosin thick filaments. Each motor has a backbone structure with a certain number of arms (N_a), and each of the arms represents 8 real myosin heads ($N_h = 8$). Thus, the total number of myosin heads represented by one motor is $N_h N_a$. The motor backbone comprises several segments with identical length and has a bare zone at its center consisting of

one or more segments. Motor arms are connected to the endpoints of the backbone segments which are not part of the bare zone. Spacing between adjacent motor arms corresponds to the length of one backbone segment between them.

The motions of the segments constituting F-actin, motor, and ACP are regulated by the Langevin equation with inertia neglected:

$$\mathbf{F}_i - \zeta_i \frac{d\mathbf{r}_i}{dt} + \mathbf{F}_i^T = 0 \quad (\text{S1})$$

where $\mathbf{r}_i$ is a position vector of the i th element, ζ_i is a drag coefficient, t is time, $\mathbf{F}_i$ is a deterministic force, and $\mathbf{F}_i^T$ is a stochastic force satisfying the fluctuation-dissipation theorem [55]:

$$\langle \mathbf{F}_i^T(t) \mathbf{F}_j^T(t) \rangle = \frac{2k_B T \zeta_i \delta_{ij}}{\Delta t} \boldsymbol{\delta} \quad (\text{S2})$$

where δ_{ij} is the Kronecker delta, $\boldsymbol{\delta}$ is a second-order tensor, and $\Delta t = 1.15 \times 10^{-5}$ s is time step. Drag coefficients are defined via an approximated form for a cylindrical object [56]:

$$\zeta_i = 3\pi\mu r_{c,i} \frac{3 + 2r_{0,i} / r_{c,i}}{5} \quad (\text{S3})$$

where μ is the viscosity of a surrounding medium, and $r_{0,i}$ and $r_{c,i}$ are the length and diameter of a segment, respectively. Position vectors of all the segments are updated at each time step via the Euler integration scheme:

$$\mathbf{r}_i(t + \Delta t) = \mathbf{r}_i(t) + \frac{d\mathbf{r}_i}{dt} \Delta t = \mathbf{r}_i(t) + \frac{1}{\zeta_i} (\mathbf{F}_i + \mathbf{F}_i^T) \Delta t \quad (\text{S4})$$

Structures of elements and deterministic forces

Deterministic forces account for i) extensional forces that maintain equilibrium lengths, ii) bending forces which maintain equilibrium angles, and iii) repulsive forces representing volume-exclusion effects between F-actins. The bending and extensional forces originate from the following potentials:

$$U_b = \frac{1}{2} \kappa_b (\theta - \theta_0)^2 \quad (\text{S5})$$

$$U_s = \frac{1}{2} \kappa_s (r - r_0)^2 \quad (\text{S6})$$

where κ_b and κ_s are bending and extensional stiffnesses, θ and θ_0 are instantaneous and equilibrium angles formed by adjacent segments, and r and r_0 are the instantaneous and equilibrium lengths of cylindrical segments, respectively. An equilibrium angle formed by two adjacent actin segments ($\theta_{0,A} = 0$ rad) and the equilibrium length of actin segments ($r_{0,A} = 140$ nm) are regulated by the bending ($\kappa_{b,A}$) and extensional ($\kappa_{s,A}$) stiffnesses of F-actin, respectively. The value of $\kappa_{b,A}$ corresponds to the persistence length of 9 μm [57]. An equilibrium angle formed by two arms of each ACP ($\theta_{0,ACP} = 0$ rad) and the equilibrium length of ACP arms ($r_{0,ACP} = 23.5$ nm) are maintained by the extensional ($\kappa_{s,ACP}$) and bending ($\kappa_{b,ACP}$) stiffnesses of ACPs, respectively.

An equilibrium angle formed by adjacent backbone segments ($\theta_{0,M} = 0$ rad) and the equilibrium length of motor backbone segments ($r_{s,M1}$) are maintained by bending ($\kappa_{b,M}$) and extensional ($\kappa_{s,M1}$) stiffnesses, respectively. The value of $\kappa_{s,M1}$ is equal to that of $\kappa_{s,A}$, whereas the value of $\kappa_{b,M}$

is larger than that of $\kappa_{b,A}$ so that the backbone does not bend significantly. The reference value of $r_{s,M1}$ is 42 nm, but it is varied in some of the simulations to increase the spacing between motor arms as well as the total length of the motor backbone. The number of segments in the bare zone determines the length of the bare zone. Two motor arms are located on each endpoint of backbone segments which are not part of the bare zone. The extension of each motor arm is regulated by the two-spring model with transverse ($\kappa_{s,M2}$) and longitudinal ($\kappa_{s,M3}$) springs. The transverse spring regulates an equilibrium distance ($r_{0,M2} = 13.5$ nm) between the endpoint of a motor backbone and an actin segment where the arm of the motor is bound, whereas the longitudinal spring maintains a right angle between the motor arm and the actin segment ($r_{0,M3} = 0$ nm).

Repulsive forces originate from a harmonic potential [58]:

$$U_r = \begin{cases} \frac{1}{2} \kappa_{r,A} (r_{12} - r_{c,A})^2 & \text{if } r_{12} < r_{c,A} \\ 0 & \text{if } r_{12} \geq r_{c,A} \end{cases} \quad (S7)$$

where $\kappa_{r,A}$ is the strength of repulsive force, and r_{12} is the minimum distance between two neighboring actin segments.

Forces exerted on actin segments by bound motor arms and ACPs or by the repulsive forces are distributed onto two ends (barbed and pointed ends) of the actin segments as described in our previous work in detail [41].

Network assembly

Unlike F-actin in bundle simulations, F-actin in network simulations is formed by stochastic processes as in our previous studies. The formation of F-actin is initiated from a nucleation event with a constant rate constant, $k_{n,A}$, with the appearance of one cylindrical segment in a random position with a random orientation perpendicular to the z direction. The polymerization of F-actin is simulated by adding cylindrical segments at the barbed end of existing filaments with a rate constant, $k_{p,A}$. The ratio of $k_{n,A}$ to $k_{p,A}$ is adjusted to result in the average filament length of ~ 10 μm . The rest of the assembly process is identical to that described in the main text.

Dynamic behaviors of ACPs

The arms of ACPs bind to binding sites located on actin segments every 7 nm without any preference of contact angle at a constant rate. In all simulations, ACPs are not allowed to unbind from F-actins after binding, forming permanent cross-links.

Dynamic behaviors of motors

The motor arms bind to binding sites on actin segments at the rate of $40N_h \text{ s}^{-1}$, where N_h is the number of myosin heads represented by each motor arm as explained above. Motor arms can bind to F-actin only when they are properly aligned with respect to the polarity of F-actin, considering that myosin heads do not interact with F-actin if the heads are inappropriately oriented [5, 59]. Note that only with the proper alignment, motor arms are able to walk toward the end of the backbone where they are connected, not toward the center of the backbone. In addition, it is assumed that two motor arms connected to the same point of the backbone are not allowed to bind to the same F-actin. After binding, motor arms walk toward the barbed end of F-actin.

The walking ($k_{w,M}$) and unbinding ($k_{u,M}$) rates of the motor arms are defined by the parallel cluster model (PCM) to mimic the mechanochemical cycle of non-muscle myosin II [60, 61]. The details of the implementation and benchmarking of the PCM in our agent-based model are extensively described in our previous study [42]. Note that $k_{w,M}$ and $k_{u,M}$ are smaller with a

higher applied load with an assumption that motor arms show catch-bond behaviors. The unloaded walking velocity and stall force of motors are set to ~ 140 nm/s and $F_{\text{st}}N_{\text{h}}$, respectively, where F_{st} is the stall force of one myosin head, ~ 5.7 pN.

Symbol	Definition	Value
$r_{0,A}$	Equilibrium length of the actin segment	1.4×10^{-7} [m]
$r_{c,A}$	Diameter of the actin segment	7.0×10^{-9} [m]
$\theta_{0,A}$	Equilibrium angle of F-actin	0 [rad]
$\kappa_{s,A}$	Extensional stiffness of F-actin	1.69×10^{-2} [N/m]
$\kappa_{b,A}$	Bending stiffness of F-actin	2.64×10^{-19} [N·m]
$r_{0,ACP}$	Equilibrium length of the ACP segment	2.35×10^{-8} [m]
$r_{c,ACP}$	Diameter of the ACP segment	1.0×10^{-8} [m]
$\theta_{0,ACP}$	Equilibrium angle of ACP	0 [rad]
$\kappa_{s,ACP}$	Extensional stiffness of the ACP	2.0×10^{-3} [N/m]
$\kappa_{b,ACP}$	Bending stiffness of ACP	1.04×10^{-19} [N·m]
$r_{0,M1}$	Equilibrium length of the motor backbone segment	42-138 [nm]
$r_{c,M}$	Diameter of the motor backbone segment	1.0×10^{-8} [N/m]
$\theta_{0,M}$	Equilibrium angle of the motor backbone	0 [rad]
$\kappa_{s,M1}$	Extensional stiffness of the motor backbone	1.69×10^{-2} [N/m]
$\kappa_{b,M}$	Bending stiffness of the motor backbone	5.07×10^{-18} [N·m]
$r_{0,M2}$	Equilibrium length 1 of the motor arm	1.35×10^{-8} [m]
$r_{0,M3}$	Equilibrium length 2 of the motor arm	0 [m]
$\kappa_{s,M2}$	Extensional stiffness 1 of a motor arm	1.0×10^{-3} [N/m]
$\kappa_{s,M3}$	Extensional stiffness 2 of a motor arm	1.0×10^{-3} [N/m]
$k_{n,A}$	Nucleation rate constant in network simulations	1.0×10^{-6} [$\mu\text{M}^{-1}\text{s}^{-1}$]
$k_{p,A}$	Polymerization rate constant in network simulations	6.0×10^4 [$\mu\text{M}^{-1}\text{s}^{-1}$]
N_h	Number of myosin heads represented by a motor arm	8
N_a	Number of arms in a single motor	4-48
$\kappa_{r,A}$	Strength of the repulsive force	1.69×10^{-3} [N/m]
N_M	Number of motors in the system	1-2,000
Δt	Time step	1.15×10^{-5} [s]
μ	Viscosity of medium	8.6×10^{-1} [kg/m·s]
$k_B T$	Thermal energy	4.142×10^{-21} [J]

Table S1.

List of parameters used in the model.

Table S2.

List of parameter values used for adopting the “parallel cluster model”.

Note that we used slightly different values for F_0 , d , and k_m from those in the literature.

Symbol	Definition	Value
k_{01}	A rate from unbound to weakly bound state	40 [s ⁻¹]
k_{10}	A rate from weakly bound to unbound state	2 [s ⁻¹]
k_{12}	A rate from weakly bound to post-power-stroke state	1,000 [s ⁻¹]
k_{21}	A rate from post-power-stroke to weakly bound state	1,000 [s ⁻¹]
k_{20}	A rate from post-power-stroke to unbound state	20 [s ⁻¹]
F_0	Force dependence	5.04×10^{-12} [N]
E_{pp}	Free energy bias toward the post-power-stroke state	-60×10^{-21} [J]
E_{ext}	External energy contribution	0 [J]
d	Step size	7×10^{-9} [m]
k_m	Spring constant of the neck linkers	1×10^{-3} [N/m]

Fig. S1.

Measurement of the bundle force.

For each cross-section located every 200 nm across the domain, segments crossing the cross-section are identified. In this example, 4 actin segments, 2 ACP segments, and 1 motor backbone segment cross the cross-section. Then, the z component of spring forces acting on those identified segments is summed as F_i . The average of F_i over all 100 cross-sections at a steady state is considered the total force acting on the bundle, F_{tot} .

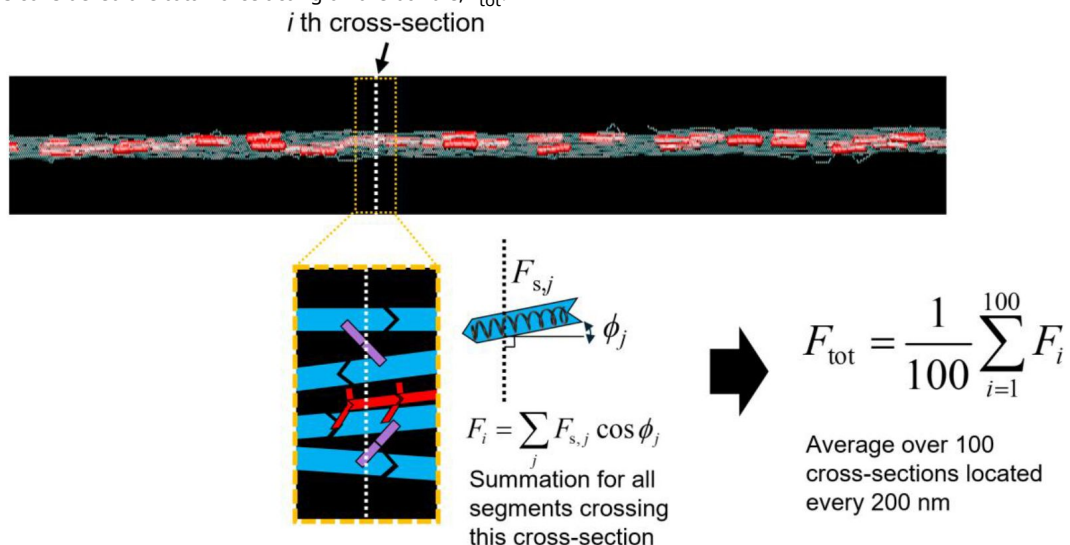


Fig. S2.

Examples of calculating the maximal number of cooperatively overlapping motors.

ξ indicates the extent of an overlap between two motors, and it is defined on the left (L) and right (R) sides of each motor. If two motors have a fully cooperative overlap, ξ is 1 on both sides of the two motors. If there is no overlap at all, ξ is 0 on both sides of the two motors. If two motors have a partially cooperative overlap, ξ is greater than 0 only on one side of the two motors, L_{ov} / L_c , where L_{ov} is an overlap distance between two motors, and L_c is the critical length required for the fully cooperative overlap. Ξ is the maximal number of cooperatively overlapping motors obtained by comparing all the combinations of motors in the system. Four examples are provided here for better understanding of this analysis method.

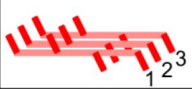
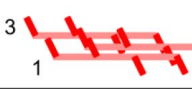
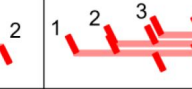
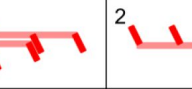
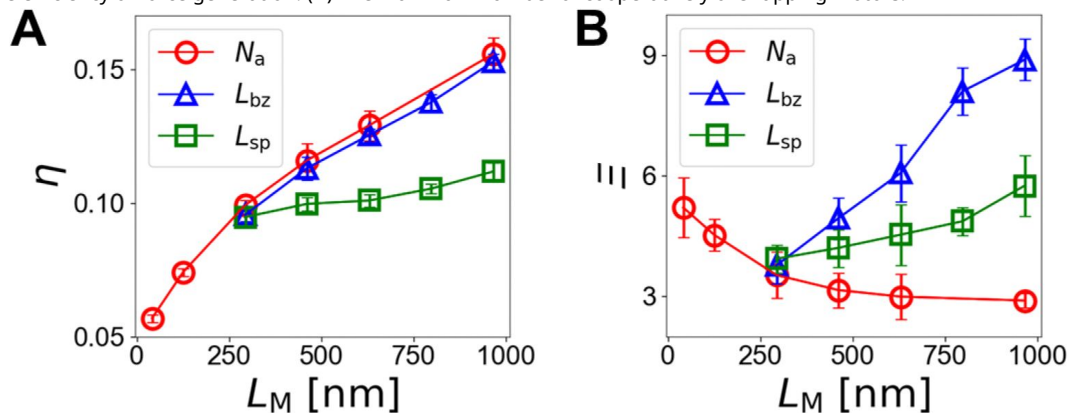
			
$\xi_{12}^L = 1, \xi_{12}^R = 1$	$\xi_{12}^L = 1, \xi_{12}^R = 1$	$\xi_{12}^L = 1, \xi_{12}^R = 1$	$\xi_{12}^L = L_{ov} / L_c, \xi_{12}^R = 0$
$\xi_{13}^L = 1, \xi_{13}^R = 1$	$\xi_{13}^L = 1, \xi_{13}^R = 1$	$\xi_{13}^L = 0, \xi_{13}^R = L_{ov} / L_c$	$\xi_{13}^L = 0, \xi_{13}^R = L_{ov} / L_c$
$\xi_{23}^L = 1, \xi_{23}^R = 1$	$\xi_{23}^L = 1, \xi_{23}^R = 1$	$\xi_{23}^L = 1, \xi_{23}^R = 1$	$\xi_{23}^L = 0, \xi_{23}^R = 0$
$\Xi = 3$	$\Xi = 3$	$\Xi = 3$	$\Xi = 1 + \max(\xi_{12}^L, \xi_{13}^R)$

Fig. S3.

Further analyses for cases with different motor length (L_M) varied by three methods.

(A) The efficiency of force generation. (B) The maximum number of cooperatively overlapping motors.



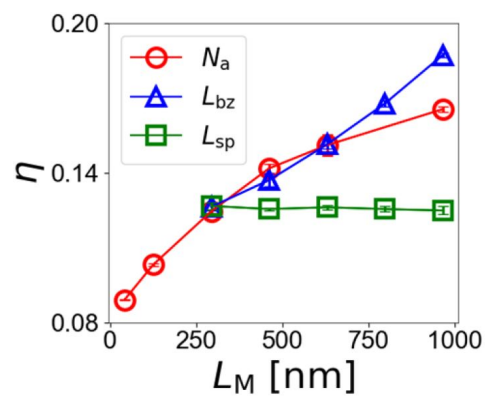


Fig. S4.

The efficiency of force generation in two-dimensional actomyosin networks with different motor length varied by three methods.

Additional information

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Reviewer #1 (Public review):

Summary:

This work by Ding et al uses agent-based simulations to explore the role of the structure of molecular motor myosin filaments in force generation in cytoskeletal structures. The focus of the study is on disordered actin bundles which can occur in the cell cytoskeleton and can be investigated with in vitro purified protein experiments. A key finding is that the force generation depends on the number of myosin motor heads and the spatial distribution of the myosin thick filaments in relation to passive crosslinkers.

Strengths:

The work develops a model where the detailed structure of the myosin motor filaments with multiple heads is represented. This allows the authors to test the dependence of myosin-generated forces on the number of myosin heads and their spatial distribution.

The work highlights that forces from multiple myosin motors within a disordered actin bundle may not simply add up, but depend on their spatial distribution in relation to passive crosslinkers.

This may explain prior experimental observations in in vitro reconstituted actomyosin bundles that the tension developed in the bundle was proportional to the number of myosin motor heads per filament rather than the number of myosin filaments. More generally, this type of modeling can guide fundamental understanding of the relationship between structure and mechanical force production.

Weaknesses:

The work focuses on the structure of myosin filaments but ignores other processes that may determine contractility of actomyosin structures such as the dynamics of crosslinker binding/unbinding and actin polymerization/depolymerization.

The authors did not vary the relative concentration of myosin motors and passive crosslinkers. This would have revealed interesting competing effects between motor and crosslink density and distribution, that their model and other studies suggest are important.

Given the above factors and the lack of direct quantitative comparisons with the experiment, the physiological significance of the work remains hard to ascertain.

<https://doi.org/10.7554/eLife.105236.2.sa2>

Reviewer #2 (Public review):

Summary:

In this study, the authors use a mechanical model to investigate how the geometry and deformations of myosin II filaments influence their force generation. They introduce a force generation efficiency that is defined as the ratio of the total generated force and the maximal force that the motors can generate. By changing the architecture of the myosin II filaments, they study the force generation efficiency in different systems: two filaments, a disorganized bundle, and a 2D network. In the simple two-filament systems, they found that in the presence of actin cross-linking proteins motors cannot add up their force because of steric

hindrances. In the disorganized bundle, the authors identified a critical overlap of motors for cooperative force generation. This overlap is also influenced by the arrangement of the motor on the filaments and influenced by the length of the bare zone between the motor heads.

Strengths:

The strength of the study is the identification of organizational principles in myosin II filaments that influence force generation. It provides a complementary mechanistic perspective on the operation of these motor filaments. The force generation efficiency and the cooperative overlap number are quantitative ways to characterize the force generation of molecular motors in clusters and between filaments. These quantities and their conceptual implications are most likely also applicable in other systems.

Weaknesses:

The detailed model that the authors present relies on over 20 numerical parameters that are listed in the supplement. Because of this vast number of parameters, it is not clear how general the findings are. On the other hand, it was not obvious how specific the model is to myosin II, meaning how well it can describe experimental findings or make measurable predictions. Although the authors partially addressed this point in the revisions, I still think it is not easy to see what are the fundamental principles that govern the behavior and how they could be different for different motor proteins.

The model seems to be quantitative, but the interpretation and connection to real experiments is rather qualitative in my point of view.

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Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

Summary:

This work by Ding et al uses agent-based simulations to explore the role of the structure of molecular motor myosin filaments in force generation in cytoskeletal structures. The focus of the study is on disordered actin bundles which can occur in the cell cytoskeleton and have also been investigated with in vitro purified protein experiments.

Strengths:

The key finding is that cooperative effects between multiple myosin filaments can enhance both total force and the efficiency of force generation (force per myosin). These trends were possible to obtain only because the detailed structure of the motor filaments with multiple heads is represented in the model.

We appreciate your comments about the strength of our study.

Weaknesses:

It is not clearly described what scientific/biological questions about cellular force production the work answers. There should be more discussion of how their simulation results compare with existing experiments or can be tested in future experiments.

Please see our response to the comment (1) below.

The model assumptions and scientific context need to be described better.

We apologize for the insufficient descriptions about the model and the scientific context. We revised the manuscript to better explain model assumptions and scientific context as described in our responses below.

The network contractility seems to be a mere appendix to the bundle contractility which is presented in much more detail.

Please see our response to the comment (6) below.

Reviewer #1 (Recommendations for the authors):

(1) It is not clearly described what scientific/biological questions about cellular force production the work answers. There should be more discussion of how their simulation results compare with existing experiments, or can be tested in future experiments. The authors do briefly mention Reference 4 where different myosin isoforms were used, but it is not clear that these experiments support the scalings predicted in this work in Figures 3-6. Also, the experiments in Ref. 4 apparently did not involve passive crosslinkers (ACPs) which are key in this study.

Thank you for the comment. In the 5th paragraph of the discussion section of the original manuscript, we applied our findings to understand how structural differences between ventral stress fibers and actin arcs could affect force generation. In addition, at the end of the discussion section, we mentioned that experiments with artificially-made myosin thick filaments could be used for verifying our results.

The experiments in Ref. 4 were only ones that we could directly compare our results with. In previous study, actomyosin bundles were experimentally created with ACPs (K.L. Weirich et al., Biophys J, 2021, 120: 1957-1970), but the motions of myosin thick filaments were only quantities measured in the experiments. In general, measuring forces generated by in vitro actomyosin bundles is very challenging. This is why the predictions from our model are particularly valuable for understanding the force generation of actomyosin structures.

(2) The architecture of the bundles seems to be prescribed by hand in these simulations. Several well-known stochastic aspects of the dynamics of actin and actin-binding proteins are not included in the model. For example, there is no remodeling of the actin structures through actin polymerization and depolymerization, or crosslink (ACP) binding and unbinding. Can the authors comment on why these effects could be neglected for the questions they want to address?

Thank you for the comment. We previously showed that the force generation process in actomyosin networks and bundles is affected by actin dynamics (Q. Yu et al., Biophys J, 2018, 115: 2003-2013) and the unbinding of ACPs (T. Kim, Biomech Model Mechanobiol, 2015, 14(2): 345-355 and W. Jung et al., Comput Part Mech, 2015, 2(4): 317-327).

However, we did not include the actin dynamics and the ACP unbinding in the current study to clearly understand the effects of the structural properties of thick filaments on the force generation process. We have learned that the stochastic behaviors of cytoskeletal components lead to noisier results, which requires us to run a much larger number of simulations to obtain statistically convincing data. We added the following paragraph in the discussion section of the revised manuscript:

“Although this study focused mainly on parameters related to motor structures, we expect that other parameters would affect the force generation process. For example, as we showed before, a decrease in ACP density would reduce forces by deteriorating connectivity between filaments. With very low ACP density, some of neighboring motors may not have ACPs between them, thus adding up their forces as shown in Fig. 2. However, such low ACP density may not maintain the structure of bundles or cross-linked networks well. In addition, the force-dependent unbinding of ACPs could change the spatial distribution of ACPs during force generation. If they behave as a slip bond which unbinds more frequently with higher forces, ACPs may not stay between two motors for long time due to high tension. Then, forces generated by two motors may have a higher chance to add up. By contrast, if they behave as a catch bond which unbinds less frequently with larger forces, more ACPs will be recruited between two motors, reducing a chance to add up

forces. The length of actin filaments is unlikely to affect the force generation process significantly unless filaments are very short. Additionally, as we showed before, actin turnover would reduce forces by competing with motor activities, change connectivity between filaments over time, and prevent motors from being stalled for long time, all of which could affect force generation.”

(3) The present study is confined to the fixed density of motors and ACPs. However, these can be easily varied in in vitro experiments. Works such as Reference 4 show an optimum in contractility vs myosin concentration. Myosins act not only to slide actin filaments but also crosslink them.

Can the authors vary myosin concentration to demonstrate such effects in their model?

As the reviewer pointed out, there is a belief that myosin thick filaments can serve as crosslinkers as well. However, unless there are a fraction of dead myosins (which remain bound on filaments without walking) or myosins dwell at the barbed ends filaments for very long time, it looks very hard for bundles or networks to generate large forces. A former experiment showed that active myosins increases the viscosity of actin networks, not elasticity (D. Humphrey et al., Nature, 2002, 416: 413-416) Computer simulations with reasonable assumptions did not show significant force generation without cross-linkers. We have tested systems with a large number of motors and a few cross-linkers in previous studies (T. Kim, Biomech Model Mechanobiol, 2015, 14(2): 345-355 and W. Jung et al., Comput Part Mech, 2015, 2(4): 317-327). We observed that large force/stress was generated momentarily, but it was relaxed very fast. It is expected that there will be similar outcomes if we try such conditions in the current study.

(4) Why is there a (factor of 1.5-2) discrepancy in the measured (F_{tot}) and estimated (F_{est}) force values in Figure 4-6? How can the authors improve their scaling arguments to capture this? What about the estimated efficiency?

Thank you for the comment. Indeed, there was a discrepancy between the actual and estimated forces. When the estimated force was calculated, we used the z positions of motors without consideration of the actual bundle geometry with multiple filaments. For example, if two motors are located on the opposite sides of the bundle (i.e., if they are located far from each other in x or y direction), forces generated by them may not counterbalance each other. Then, the estimated force can be smaller than the actual force because counterbalance between motors can be overcounted. The original manuscript had the following sentences to clarify this point: “ F_{est} was generally smaller than F_{tot} because this analysis does not account for actual bundle geometry consisting of multiple F-actins; if two motors are located far from each other in x or y direction, they may not counterbalance or add up forces. Nevertheless, we found that F_{est} captures the overall dependence of F_{tot} on parameters well.”

(5) Several choices of parameter values used in the simulations are not clear:

a) Why consider F actin of 140 nm specifically? Actin can come in a range of lengths. How do their results depend upon the length scale of actin?

It seems that there is a misunderstanding. 140 nm is the equilibrium length of one actin segment in our model. The actual F-actin consists of multiple actin segments. The length of F-actin was 9 μm in bundle simulations and 10 μm (average) in network simulations. We expect that the general tendency of our results would not change with different filament length. However, if filament length becomes too short, the force generation process would be impaired due to lack of connectivity between filaments.

b) Similarly, very specific values of myosin backbone length (42 nm), number of myosin heads (8), number of arms (24), and Actin Cross-linking Proteins (ACPs). What informs these values and how will the results change if they are different? It is not especially clear how an "Arm" differs from "heads" and what kind of coarse-graining is involved.

In the "model overview" section of the original manuscript, we mentioned the following to clarify the definitions of motor arms and motor heads:

"To mimic the structure of bipolar filaments, each motor has a backbone, consisting of serially linked segments, and two arms on each endpoint of the backbone segments that represent 8 myosin heads ($N_h = 8$)."

We devised this coarse-graining scheme of myosin thick filaments in our previous work (T. Kim, *Biomech Model Mechanobiol*, 2015, 14(5): 1143-1155). Through extensive tests, we showed that force generation and motor behaviors are largely independent of coarse-graining level. In other words, a motor with the same value of $N_h N_a$ leads to similar outcomes regardless of the value of N_a . However, in a bundle with multiple filaments, each motor has a sufficient number of arms to ensure simultaneous interactions with those filaments. This is why we decided to use $N_h = 8$ and $N_a = 24$.

To match the length of thick filaments and the total number of heads ($N_h N_a$) in the model with real myosin thick filaments, we have used 42 nm for each backbone length. Varying this length is equivalent to a variation in L_{sp} that we did for Fig. 6.

We used high ACP density to ensure connections between all neighboring pairs of actin filaments. We already showed how the presence of ACPs affects the force generation process in Fig. 2 using two actin filaments. It is expected that a variation of ACP density would affect our results to some extent. Since the main focus of the current study is the structural properties of motors, we did not explore the effects of ACP density. I hope that the reviewer would understand our intention.

(6) The manuscript focuses on disordered bundles with only one figure on networks. However, actin fibers also ubiquitously exist as disordered networks, and it is important to explore in more detail the contractile forces in such network arrangements.

We appreciate the comment. Because we plan to delve into the effects of motor structures on the force generation in networks as a follow-up study, we showed the minimal results in the current study to prove the generality of our findings. I hope that the reviewer would understand our intention and plan.

It is not described very clearly how these networks were generated.

We apologize for lack of explanation about how the networks were generated. We added the following section in Supplementary Text of the revised manuscript:

“Network assembly

Unlike F-actin in bundle simulations, F-actin in network simulations is formed by stochastic processes as in our previous studies. The formation of F-actin is initiated from a nucleation event with a constant rate constant, $k_{n,A}$, with the appearance of one cylindrical segment in a random position with a random orientation perpendicular to the z direction. The polymerization of F-actin is simulated by adding cylindrical segments at the barbed end of existing filaments with a rate constant, $k_{p,A}$. The ratio of $k_{n,A}$ to $k_{p,A}$ is adjusted to result in the average filament length of $\sim 10 \mu\text{m}$. The rest of the assembly process is identical to that described in the main text.”

Crosslinked biopolymers like actin typically form disordered elastic networks with their coordination number below rigidity percolation threshold ($z=4$ in 2D), see for example review by Broedersz and Mackintosh Rev. Mod. Phys. 2013. Such networks should exist in the bending-dominated regime, where bending forces play a vital role in force propagation. Was that observed in the simulations? Why or why not?

We appreciate the comment. We are aware of the bending-dominated regime and indeed showed the importance of the bending stiffness of actin filaments at low shear strain level in our previous work (T. Kim et al., PLOS Comput Biol, 2009, 5(7): e1000439). In case of active networks with motors, such a bending-dominated regime has not been observed without external shear strain. Instead, buckling of actin filaments was found to be essential for breaking symmetry between tensile and compressive forces developed by motor activities. We have shown that the free contraction of networks is inhibited if filament bending stiffness is increased substantially (J. Li et al., Soft Matter, 2017, 13: 3213-3220 and T. Bidone et al., PLOS Comput Biol, 2017, 13(1): e1005277). We expect that contractile forces generated by bundles or networks will be reduced significantly if we highly increase bending stiffness. However, considering the focus of the current study is on the structural properties of motors, we did not perform such simulations.

(7) It would be interesting to see the simulated predictions of the bundle or network contraction dynamics. This can be done by changing to free boundary conditions so that the bundle can contract.

Thank you for the suggestion. We have previously investigated the free contraction of actomyosin networks with different motor density and ACP density (J Li et al., Soft Matter, 2017, 13: 3213). We observed that the rate of network contraction was higher with more motors and ACPs. However, we did not test the effects of the structural properties of thick filaments in the previous study. We plan to investigate the effects in future studies because the focus of the current study is the force generation process. Please note that in the discussion section of the original manuscript, we mentioned the following:

“Although we focused on force generation, the contractile behaviors of actomyosin structures (i.e., a decrease in length) have also been of great interest. Our model can be used to study such contractile behaviors by deactivating the periodic boundary condition and removing connection between one end of bundle/network and a domain boundary as done previously [20]. To achieve higher contractile speed with the same total number of myosin heads, the existence of multiple contractile units would be better as suggested in a previous work [4]. This means that there is a trade-off between force generation and contractile speed. Previous studies also showed that the contractile speed of networks is proportional to motor density [18, 43, 51]. We may be able to use our model to systematically investigate how the contractile

speed is regulated by parameters that we tested in this study, including the number, distribution, length, and structure of motors.”

Minor suggestions for improvement:

(1) What are the vertical markers in Figures 1E and F? They should be labelled. if they are crosslinkers, it is not clear why the color is different from Figure 1A and B.

We believe that the reviewer meant Figs. 2E, F. Those vertical lines are indeed ACPs (crosslinkers). We changed the color of ACPs in Fig. 1A and Fig. 2B-D to purple to be consistent. In addition, we changed the colors of two filaments in Figs. 2B-D slightly to be consistent with Fig. 2E.

(2) To help understanding, please include a figure showing how forces are measured.

We added Fig. S1 in the revised manuscript to explain how the bundle force is calculated.

(3) It should be possible to extend the scaling arguments to predict what is the crossover myosin density (N_M) in Figure 4a at which the efficiency changes from going as $1/N_M$ to saturating.

As the reviewer might have observed, the slope of the efficiency in Fig. 4A gradually changes, rather than showing a sharp transition. Thus, it is hard to define one crossover myosin density.

Similarly, what are the slopes in Figure 6a-b?

We drew the reference lines in those two plots. Unfortunately, we do not have explanations about the origin of these slopes.

(4) Some more explanation for the observed values should be added. Figure 4: Why does efficiency plateau at a value close to 0.8 in (A)?

We assume that the reviewer meant the plateau of η close to 0.08, not 0.8. Our speculation for the origin of this plateau value is related to L_M (= 462 nm under the reference condition). Ideally, ~43 motors are required to cover the entire length of the bundle (= 20 μ m). Under this condition, η is ~0.023. Although this is not 0.08, we believe that these two values are related to each other. For example, if we increase L_M , this plateau level would increase. We added the following sentences in the result section of the revised manuscript:

“The plateau level of η at ~0.08 is related to the minimum number of motors required for saturating an entire bundle, implying that the plateau level would be higher if each motor is longer.”

Figure 5: Overlapping between motors seems to increase the total force applied by them because of cooperative effects. However, it is not abundantly clear why that should peak at a value of $f = 0.06$.

As shown in Fig. 5B, smaller f always results in higher F_{tot} due to higher level of cooperative overlap. The minimum value of f we tested in this study was 0.06, so F_{tot} was maximal at $f = 0.06$.

(5) Why is the network force expected to scale approximately as $\sqrt{N_M}$? Is it because of the 2D geometry where the number of motors along the x or y-direction scale as $\sqrt{N_M}$?

We initially thought that the weaker dependence of the total force on N_M was related to the random orientations of motors. However, if the network is fully saturated with motors, the inclusion of more motors will increase forces in both x and y directions almost linearly, resulting in the direct proportionality of F_{tot} to N_M . Our new hypothesis for weaker dependence is consistent with the reviewer's speculation; the network is not fully saturated even with 1000 motors, so the entire regime shown in Fig. 7B corresponds to that with $N_M < 100$ in Fig. 4A where similar weaker dependence on N_M was observed. We added the following sentence in the result section of the revised manuscript to clarify this point:

“the average number of motors in each direction which can experience the cooperative overlap would be $\sim \sqrt{N_M}$. Maximal N_M tested with the network was $\sim 2,500$, so the dependence of F_{tot} on N_M with the network is similar to that with $N_M < \sim 50$ with the bundle (Fig. 4A).”

(6) Figures 6 D and A: Figure 6D suggests that there is a more full overlap in the cases where there was a longer bare zone or larger spacing between motor arms. However, the quantification of the total force in A shows that the force is highest for the case where LM was increased by increasing the number of arms. Why do the authors think that is? I would expect from the explanation in Fig 6D that the L_{sp} and L_{bz} would be higher than N_a in Fig 6A.

Fig. 6D shows a difference in the level of the cooperative overlap (Fig. 6D) between two motors. As

the reviewer pointed out, the case with more arms shows the lowest Fig. 6D, resulting in the

lowest Fig. 6D as we showed in Fig. S2B. However, as shown in Eq. 7, the total force is a function

of both N_a and Fig. 6D. Thus, due to higher N_a and lower Fig. 6D, the force in the case with different N_a

can be similar to that in the case with different L_{bz} . In the original manuscript, we had the following sentence to explain how the force can be similar between the two cases:

“Thus, Fig. 6D was higher (Fig. S2B, blue), resulting in higher F_{tot} and η despite smaller N_a .”

Reviewer #2 (Public review):

Summary:

In this study, the authors use a mechanical model to investigate how the geometry and deformations of myosin II filaments influence their force generation. They introduce a force generation efficiency that is defined as the ratio of the total generated force and the maximal force that the motors can generate. By changing the architecture of the myosin II filaments, they study the force generation efficiency in different systems: two filaments, a disorganized bundle, and a 2D network. In the simple two-filament systems, they found that in the presence of actin crosslinking proteins motors cannot add up their force because of steric hindrances. In the disorganized bundle, the authors identified a critical overlap of motors for cooperative force generation. This overlap is also influenced by the arrangement of the motor on the filaments and influenced by the length of the bare zone between the motor heads.

Strengths:

The strength of the study is the identification of organizational principles in myosin II filaments that influence force generation. It provides a complementary mechanistic perspective on the operation of these motor filaments. The force generation efficiency and the cooperative overlap number are quantitative ways to characterize the force generation of molecular motors in clusters and between filaments. These quantities and their conceptual implications are most likely also applicable in other systems.

Thank you for the comments about the strength of our study.

Weaknesses:

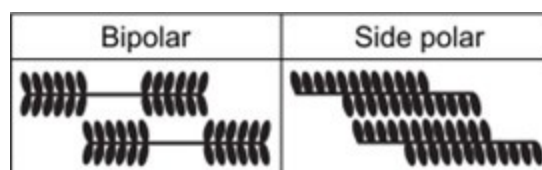
The detailed model that the authors present relies on over 20 numerical parameters that are listed in the supplement. Because of this vast amount of parameters, it is not clear how general the findings are. On the other hand, it was not obvious how specific the model is to myosin II, meaning how well it can describe experimental findings or make measurable predictions. The model seems to be quantitative, but the interpretation and connection to real experiments are rather qualitative in my point of view.

As the reviewer mentioned, all agent-based computational models for simulating the actin cytoskeleton are inevitably involved with such a large number of parameters. Some of the parameter values are not known well, so we have tuned our parameter values carefully by comparing our results with experimental observations in our previous studies since 2009. We were aware of the importance of rigorous representation of unbinding and walking rates of myosin motors, so we implemented the parallel cluster model, which can predict those rates with consideration of the mechanochemical rates of myosin II, into our model. Thus, we are convincing that our motors represent myosin II.

In our manuscript, our results were compared with prior observations in Ref. 4 (Thoresen et al., Biophys J, 2013) several times. In particular, larger force generation with more myosin heads per thick filament was consistent between the experiment and our simulations.

Our study can make various predictions. First, our study explains why non-muscle myosin II in stress fibers shows focal distributions rather than uniform distributions; if they stay closely, they can generate much larger forces in the stress fibers via the cooperative overlap. Our study also predicts a difference between bipolar structures (found in skeletal muscle myosins and nonmuscle myosins) and side polar structures (found in smooth muscle myosins) in terms of the likelihood of the cooperative overlap. As shown below, myosin filaments with the bipolar structure can add up their forces better than those with the side polar structure when their overlap level is the same.

Author response image 1.



It was often difficult for me to follow what parameters were changed and what parameters were set to what numerical values when inspecting the curve shown in the figures. The manuscript could be more specific by explicitly giving numbers. For example, in the caption for Figure 6, instead of saying "is varied by changing the number of motor

arms, the bare zone length, the spacing between motor arms", the authors could be more specific and give the ranges: "is varied by changing the number of motor arms from ... to .., the bare zone length from .. to..., and the spacing between motor arms from .. to ..".

This unspecificity is also reflected in the text: "We ran simulations with a variation in either L_{sp} or L_{bz} " What is the range of this variation? "When L_M was similar" similar to what? "despite different N_M ." What are the different values for N_M ? These are only a few examples that show that the text could be way more specific and quantitative instead of qualitative descriptions.

We appreciate the comment. In the revised manuscript, we specified the range of the variation in each parameter.

In the text, after equation (2) the authors discuss assumptions about the binding of the motor to the actin filament. I think these model-related assumptions and explanations should be discussed not in the results section but rather in the "model overview" section.

Thank you for pointing this out. In the original manuscript, we described all the details of the model in Supplementary Material. We feel that the assumptions about interactions between motors and actin filaments are too detailed information to be included in the model overview section.

The lines with different colors in Figure 2A are not explained. What systems and parameters do they represent?

The different colors used in Fig. 2A were used for distinguishing 20 cases. We added the explanation about the colors in the figure caption in the revised manuscript.

Reviewer #2 (Recommendations for the authors):

To guarantee the reproducibility of the results, I recommend that the authors publish their simulation code on GitHub.

We appreciate the reviewer's suggestion. Following the suggestion, we prepared and posted the code on GitHub as mentioned in the Data Availability of the revised manuscript: The source code of our model is available on GitHub: <https://github.com/ktyman2/ThickFilament>

<https://doi.org/10.7554/eLife.105236.2.sa0>